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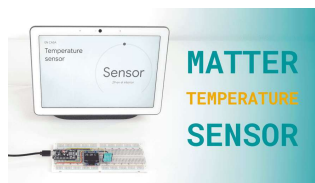
Matter Temperature Sensor with the Arduino Nano Matter

Learn how to build a Matter temperature sensor using the Arduino Nano Matter

Author · Christopher Méndez · Last revision · 02/17/2021

Overview

This tutorial will teach you how to use the Arduino Nano Matter to create a Matter sensor to monitor your room or workplace temperature.



Temperature sensor overview

Thanks to the seamless compatibility of the Nano Matter with almost any Matter network we can easily integrate our sensor with Amazon Alexa, Google Assistant, Apple HomeKit, Home Assistant and even any professional custom solution.

We have prepared a short demo in video format in case you are a visual learner.

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Build a Matter-Enabled
Temperature Sensor with the
Arduino Nano Matter



Hardware and Software Requirements

Hardware Requirements

[Arduino Nano Matter](#) (x1)

DHT11 Temperature and
Humidity sensor (x1)

I2C OLED Display SSD1306 (x1)

Breadboard (x1)

Jumper wires

Google Nest Hub Max (Thread
Border Router) (x1)

[USB-C® cable](#) (x1)

Software Requirements

[Arduino IDE 2.0+](#) or [Arduino
Cloud Editor](#)

[Google Home App](#)

U8g2 library to control the OLED display. You can install it from the Arduino IDE library manager.

DHT library. Download from this [branch](#) so it supports the Nano Matter.

Download the Project Code

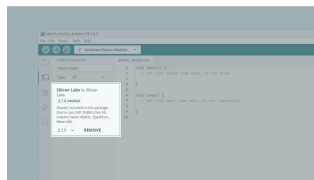
DOWNLOAD THE CODE



Download the complete project code [here](#).

Board Core and Libraries

The **Silicon Labs** core contains the libraries and examples you need to work with the board's components, such as its Matter, Bluetooth® Low Energy, and I/Os. To install the Nano Matter core, navigate to **Tools > Board > Boards Manager** or click the Boards Manager icon in the left tab of the IDE. In the Boards Manager tab, search for `Nano Matter` and install the latest `Silicon Labs` core version.

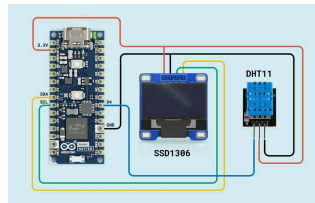


Installing the Silicon Labs core in the Arduino IDE

Project Setup

Schematic Diagram

Use the following connection diagram for the project:

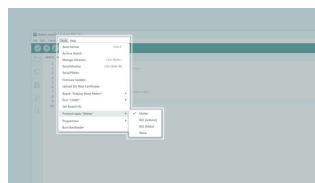


Project wiring diagram

The OLED display and the DHT temperature sensor are powered by the Nano Matter `3.3V` output pin. The OLED display uses I2C for communication and the DHT sensor uses a one-wire digital communication connected to `D4`.

Programming

In the Arduino IDE upper menu, navigate to **Tools > Protocol stack** and select **Matter**.



Matter Protocol stack selected

Copy and paste the following sketch

```

1  #include <Matter.h>
2  #include <MatterTemperature.h>
3  #include "DHT.h"
4  #include <U8x8lib.h>
5  #include <Wire.h>
6
7  U8X8_SSD1306_128X64_NONAME_F_HW_I2C u8x8(U2R_I2C_ADDRESS, U2R_I2C_CLK, U2R_I2C_DATA);
8
9  MatterTemperature matterTempSensor;
10
11 #define DHTPIN D4 // Labeled D4 on the Nano Matter
12
13 #define DHTTYPE DHT11
14
15 DHT dht(DHTPIN, DHTTYPE);
16
17 float temp = 0;
18
19 void setup() {
20   Serial.begin(115200);
21   Matter.begin();
22   matterTempSensor.begin();
23   dht.begin();
24
25   u8x8.begin();
26   u8x8.setFont(u8x8_font_4x8);
27
28   pinMode(BTN_BUILTIN, INPUT_PULLUP);

```

The structure of this example code is very simple, the main functions are explained below:

In the `setup()` function we initialize the Matter connectivity, the OLED display and temperature sensor.

In the `loop()` function we measure the temperature and update the device value to be monitored in the Matter network.

Once you uploaded the example code to the Nano Matter, open the Serial Monitor and reset the board.

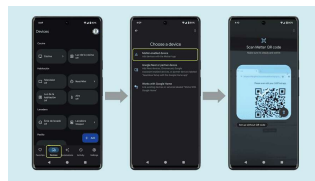


After the reset you will find on the serial port the URL that generates the QR for the Matter device commissioning.

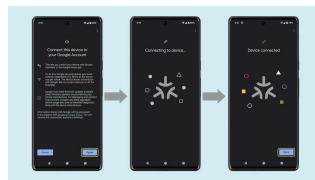
Adding the Device (Commissioning)

Copy and paste the QR code URL on your favorite web browser and a unique QR code will be generated for your board.

Go to your **Google Home** app, navigate to **devices** and tap on **Add**, select the **Matter-enabled device** option and scan the QR code.



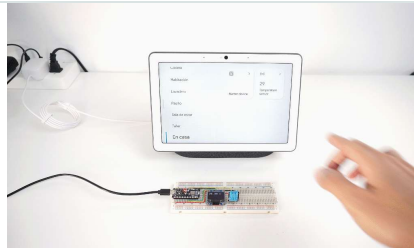
Adding the device to
Google Home app



Your device is successfully
added

Final Results

Finally, you will be able to monitor your room temperature from your smartphone, hub or asking your personal assistant.



You can also see the temperature value on the device OLED display.

Conclusion

In this tutorial we have learned how to create a Matter enabled temperature sensor that can be monitored from our smartphone and personal assistant. The Nano Matter allows us to seamlessly integrate our own sensor as a commercial product with our current smart home ecosystem.

Next Steps

You can take this solution even further by adding the humidity measuring capability of the DHT11 sensor and integrate it as a second sensor to your Matter network.

Suggest changes	Need support?	License
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